

Red Basin (Desert / Reconnaissance & Long-Range Warfare) Map Design

Executive Summary

Red Basin is a vast desert plateau battlefield built for long-range engagements, reconnaissance, and sensor-driven tactics. The terrain features wide open basins, sandstone ridges, dry riverbeds, abandoned industrial pump stations, and sparse cover. The map emphasizes Obsidian Protocol's autonomous scouting behaviors, long-range sensor warfare, comms-relay networks, and logistics stretched across large open distances. In AR (Real Light/War), the map anchors at ~4-5 m viewing distance, covering a 10x10 m physical area representing "1 km" of desert terrain. Key features include elevated mesas for origins and sensors, dry canyons for flanking, industrial pump stations, and critical mesa hubs.

Design Goals & Gameplay

Long-Range Desert Combat

Red Basin prioritizes long-range engagements. Open sightlines allow sensors, artillery, and recon drones to dominate. Sparse cover forces careful movement and positioning.

Autonomy & Recon Behavior

Autonomous AI scouts ahead, uses ridges for vantage, and avoids exposed basins. Units adapt to shifting sandstorms and sensor disruptions.

Sensor & Information Warfare

Heat distortions, dust, and mirage effects degrade sensors. Relay towers on mesas restore clarity. Destroying relays creates massive fog-of-war zones across the basin.

Logistics & Resources

Sparse resources (small cists) long open routes vulnerable to ambush. Controlling the Pump Station and Desert Depot provides major resupply advantages.

Deployment & Progress

A hard deployment budget (~10,000 points) forces players to choose between long-range units or fast recon vehicles. Blue deploys from the southwest canyon mouth, Red deploys from the northeast plateau.

Spatial Constraints (Nreal AR)

Device Capabilities

Target Lightbox: "10" diagonal FOV, 1800x1800 per-eye resolution. Optimal viewing distance ~4-5 m.

Playable Area & Scaling

10x10 m physical area = 1 km² desert basin (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion

Horizon projection across 1000+ horizontal planes. Depth-cue (if available) provides limited occlusion. Virtual objects render over real-world view, assume minimal real-world occlusion.

Layout Overview

Coordinate System

1000x1000 grid (1 Unit = 1 m). Origin (0,0) = southwest corner.

Key Locations

Blue HQ (Canyon Mouth) — (0,0). 120x120 m fortified canyon entrance with recon vehicles.

Red HQ (Nreal Plateau) — (1000,1000). 120x120 m fortified sandstone plateau with comms tower.

Central Basin — (500,500). 100x100 m wide open desert floor: long-range combat zone.

Desert Depot — (1200,800). 50x50 m supply crates. Fuel drums, recon drones.

Industrial Pump Station — (800,100). 120x120 m Pump, storage, water tanks, abandoned machinery.

Dry Riverbed — (0,400) to (400,1000). Shallow canyon, eroded bank of cover.

Mesa A — (100,150). +200 m Scarer and sensor vantage point.

Mesa B — (1100,600). +100 m Note tower and sensor hub.

Canyon Pass — (100,0) to (400,100). Narrow, Redline route with limited cover.

Neutral Forward Outpost — (400,100). Sensor research zone.

Chokepoints & Sightlines

Canyon Pass

Narrow, eroded creek ideal for ambush.

Dry Riverbed

Provides partial cover, key flanking path.

Open Basin

Exposes long-range sightlines; artillery and sensors dominate.

Sightlines

Mesa A provides long-range sensor coverage (~100 m). Heat distortion and dust storms create fog-of-war pockets.

Deployment Zones

Blue Deployment: NE canyon mouth + near rally point.

Red Deployment: NE plateau + near rally point.

Neutral Forward Zone: Outpost at (100,100) for drone drops.

Assets & Environment

Terrain

Sandstone cliffs, dunes, dry riverbeds, mesas, desert rock formations.

Vegetation

Sparse shrubs, dry grass patches, small desert trees.

Buildings & Props

Industrial: Pumps, tanks, pipes, generators. Military: Watchtowers, antennas, sandbag positions. Logistics: Trucks, crates, fuel drums, recon drones.

Effects

Dust storms, heat shimmer, smoke plumes, wind gusts.

Audio

Wind, distant machinery, radio chatter, sand movement.

LOD & Budget

2048 for terrain/buildings, 1024 for props, 512x15 compression. Target 1200 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal Architecture + real SDK (NREAL). ARKit/ARCore enabled.

Rendering

Temporal render over, stationary lights, baked GI, aggressive culling.

Markers

Single radial anchor for entire map.

Interaction

Phone gesture for pan/zoom, HUD on screen space widgets.

Multiplayer

Client-server architecture, offload AI/physics to server.

Implementation Plan & Testing

Milestones

- Prototype Layout — Month 1 Block out mesas, basin, canyon, AR anchor test.
- Core Systems — Month 2 Foundation, sensor base, autonomous behaviors.
- AI Interaction — Month 3 Anchors, vision detection, basic interactivity.
- Content Fill — Month 4-5 Final models, textures, lighting, audio.
- Gameplay Polish — Month 6-7 Optimization, balance, VFX (dust storms, heat shimmer).
- Testing & CI — Month 8-10 Multiplayer stress test, sensor validation.

Testing Plan

Autonomy

Verify scout behavior, long-range targeting, pathfinding.

Comms/Sensors

Test relay distribution, fog-of-war updates, LOS checks.

Performance

400 FPS AR render on target device.

QA Checklist

Navigation, cover usage, deployment budget enforcement, anchor drift, sensor accuracy.